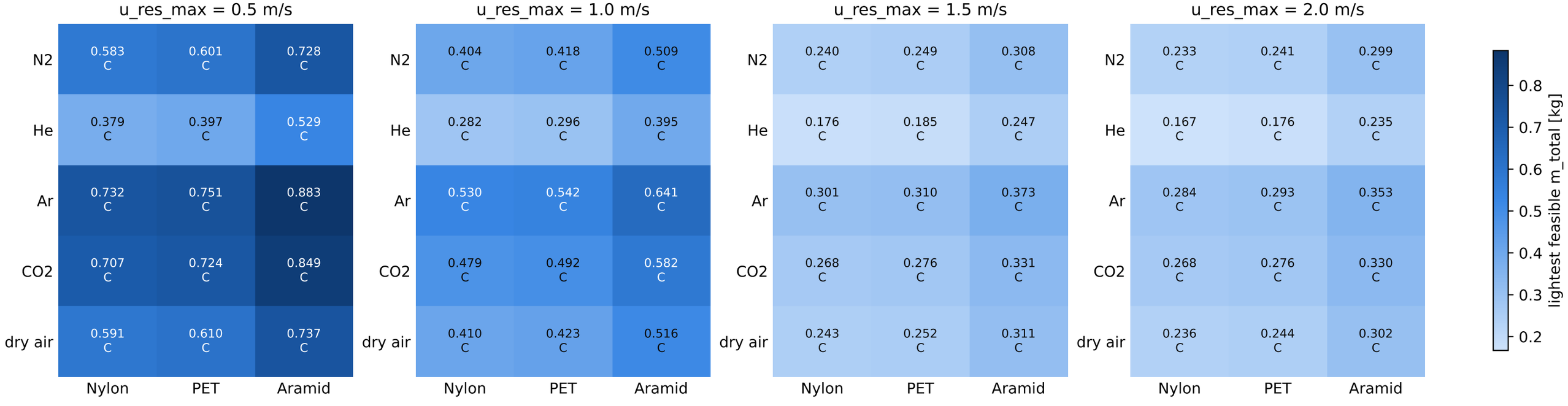
Lightest full-fidelity feasible mass per gas x material (C = cylinder, S = sphere, — = no feasible design)
u_res_max = 0.5 m/s
u_res_max = 1.0 m/s
u_res_max = 1.5 m/s
u_res_max = 2.0 m/s
N2
He
Ar
CO2
dry air
Nylon
PET
Aramid
0.583 C
0.601 C
0.728 C
0.379 C
0.397 C
0.529 C
0.732 C
0.751 C
0.883 C
0.707 C
0.724 C
0.849 C
0.591 C
0.610 C
0.737 C
0.404 C
0.418 C
0.509 C
0.282 C
0.296 C
0.395 C
0.530 C
0.542 C
0.641 C
0.479 C
0.492 C
0.582 C
0.410 C
0.423 C
0.516 C
0.240 C
0.249 C
0.308 C
0.176 C
0.185 C
0.247 C
0.301 C
0.310 C
0.373 C
0.268 C
0.276 C
0.331 C
0.243 C
0.252 C
0.311 C
0.233 C
0.241 C
0.299 C
0.167 C
0.176 C
0.235 C
0.284 C
0.293 C
0.353 C
0.268 C
0.276 C
0.330 C
0.236 C
0.244 C
0.302 C
lightest feasible m_total [kg]
0.8
0.7
0.6
0.5
0.4
0.3
0.2